

FAKE NEWS DETECTION SYSTEM

Project Report and Technical Documentation

Abstract

The Fake News Detection System is a full-stack web application that classifies submitted news as Real or Fake using TF-IDF text features and a supervised machine learning classifier. It combines a React/Vite frontend with a Django REST backend, JWT authentication, prediction history, analytics, exports and an administrator dashboard.

Objectives

Develop a web-based classifier; provide secure authentication; separate normal-user and admin functionality; integrate the trained ML model; maintain prediction history; provide charts and exports; and deliver a responsive user interface.

Technology Stack

React.js, Vite, CSS, Chart.js, Recharts, Django, Django REST Framework, Simple JWT, pandas, scikit-learn and joblib.

Machine Learning

Fake.csv and True.csv are labeled Fake (0) and Real (1). Title and text are combined, cleaned, transformed with TF-IDF, and classified using PassiveAggressiveClassifier. Model artifacts are stored with joblib. The development experiment achieved approximately 99.47% observed accuracy.

System Architecture

React UI → Axios client → Django REST API → preprocessing → TF-IDF vectorizer → classifier → prediction/history database → dashboard analytics.

Main Features

Registration and login, JWT authentication, normal-user dashboard, admin dashboard, prediction, result card, statistics, pie chart, trend chart, history search/filter, delete/clear history and PDF/CSV export.

Testing

Registration, login, invalid login, prediction, history, deletion, clearing, charts, exports and protected admin access were tested successfully during development.

Future Enhancements

Update the deprecated classifier, calibrate confidence, add richer evaluation metrics, multilingual support, external fact checking, production deployment and automated tests.

Conclusion

The project demonstrates a complete end-to-end machine-learning web application, connecting model training and prediction with authentication, persistence, analytics and administration.